

1. Administrative & Study Identification

Full Study Title: A Phase I, Open-label, Fixed-sequence Study to Evaluate the Effect of Ceralasertib on Pharmacokinetics of Drug X, Drug Y and Drug Z in Participants With Advanced Solid Tumours
Short Title/Acronym: A study to evaluate the effect of ceralasertib on Drug X, Drug Y and Drug Z
Protocol Number: D533BC00003
NCT Number: NCT06929260
Sponsor Name: AstraZeneca
Investigational Product: Ceralasertib (AZD6738), Drug X, Drug Y, Drug Z
Phase of Development: Phase I
Medical Monitor: AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective
Primary Objective: To assess the effect of ceralasertib on the pharmacokinetics (PK) of Drug X, Drug Y and Drug Z in participants with advanced solid tumours.
Secondary Objectives: To evaluate the safety, tolerability, and steady-state pharmacokinetic parameters of ceralasertib when administered in a fixed sequence with Drug X, Y, and Z.

2.2 Background & Rationale
To evaluate the potential pharmacokinetic drug-drug interactions between ceralasertib (an orally active ATR kinase inhibitor) and Drug X, Drug Y, and Drug Z. This Phase I study was designed to investigate these interactions in patients with advanced solid tumours to ensure safe co-administration profiles for future clinical applications.
(Note: The study was terminated early because the clinical development programme for Ceralasertib was discontinued).

3. Study Design & Methodology

3.1 Design Framework
Study Type: Interventional
Control Type: Single-arm (3-period fixed-sequence study)
Blinding: Open-label
Randomization: N/A (Fixed-sequence)

3.2 Planned Enrollment & Duration
Target \$N\$: 1 participant (Originally planned for 20; revised due to early termination)
Number of Sites: Multiple sites (Europe)
Estimated Study Start: 21-May-2025
Estimated Primary Completion: 15-Oct-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically documented locally advanced or metastatic solid tumour(s) of non-small cell lung cancer, ovarian cancer, endometrial cancer, breast cancer or prostate cancer at Screening. 2. Eastern Cooperative Oncology Group Performance Status (ECOG PS) of 0 or 1 with no deterioration over the 2 weeks prior to dosing. 3. Ability to swallow and retain oral medication. 4. Minimum life expectancy ≥ 12 weeks in the opinion of the Investigator. 5. Adequate organ and marrow function during Screening, body weight > 30 kg, and no cancer-associated cachexia.

4.2 Exclusion Criteria 1. Spinal cord compression or brain metastases for at least 4 weeks prior to start of study intervention unless asymptomatic and stable. 2. Any prior treatment with an ATR inhibitor or checkpoint kinase inhibitor, or diagnosis of ataxia telangiectasia (ATR). 3. Participants with confirmed severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) or active infection requiring systemic antibiotics, antifungal or antiviral drugs. 4. Refractory nausea and vomiting, chronic gastrointestinal diseases associated with diarrhoea, or previous significant bowel resection.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants receive ceralasertib twice daily (BD) from Day 1 to Day 7. Participants also receive a single dose of Drug X on Day 5 and Day 22. Similarly, a single dose of Drug Y and Drug Z is administered on Day 7 and Day 28. **Route of Administration:** Oral **Duration of Treatment:** 28 days (across 3 periods) plus follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care therapies that do not conflict with the PK assessments. **Prohibited:** Concomitant treatment of central nervous system depressants, opioids, and centrally acting anti-hypertensive agents. Proton pump inhibitors, histamine H2 receptor antagonists, and other anti-acid agents are also excluded.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening		Visit 1 Informed Consent, Medical History, Organ/Marrow Function Labs, ECG
Period 1	Visit 2 (Days 1-7)	Ceralasertib BD dosing; Drug X single dose (Day 5); Drug Y & Z single dose (Day 7)
Period 2	Visit 3 (Day 22)	Wash-out (≥ 48 hours); Drug X single dose
Period 3	Visit 4 (Day 28)	Wash-out (≥ 48 hours); Drug Y & Drug Z single dose
End of Study	Visit 5	Follow-up: Safety assessments, final PK blood draws, AE monitoring

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Effect of ceralasertib on the pharmacokinetics (PK) of Drug X, Drug Y, and Drug Z. **Measurement Method:** Serial blood sampling and central laboratory PK concentration analysis.

7.2 Secondary & Exploratory Endpoints Evaluation of overall safety and tolerability (Adverse Events [AEs]). Pharmacokinetic profiling of Ceralasertib at steady state.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of AEs throughout the dosing periods and follow-up, specifically tracking gastrointestinal toxicities and resting ECG abnormalities. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline for any SAEs or Grade 3/4 toxicities. **Data Monitoring Committee (DMC):** Internal safety review overseen by the Sponsor.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Analysis Set and Safety Analysis Set. **Sample Size Justification:** Study originally powered for $N=20$ to detect meaningful PK differences. The study was ultimately terminated early with actual $N=1$. **Handling of Missing Data:** Standard PK non-compartmental analysis rules applied; however, statistical analyses were truncated due to early termination.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Intensive onsite monitoring consistent with Phase I PK study requirements. **Audit Trail:** Source data verification and eCRF locking managed through AstraZeneca quality protocols.

11. Study Identifiers & Governance

Primary ID: D533BC00003 **Registry Number:** NCT06929260 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** A study to evaluate the effect of ceralasertib on Drug X, Drug Y and Drug Z **Official Regulatory Title:** A Phase I, Open-label, Fixed-sequence Study to

12. Clinical Framework (Oncology PK Specific)

Primary Condition: Advanced Solid Tumours (NSCLC, Ovarian, Endometrial, Breast, Prostate) **Diagnosis Threshold:** Histologically or cytologically documented locally advanced or metastatic solid tumour(s) **Medical Methodology:** Fixed-sequence pharmacokinetic (PK) drug-drug interaction evaluation **Optimization Target:** Characterizing the PK profile of Drug X, Y, and Z when co-administered with an ATR inhibitor

13. Study Procedures & Methodology

Management Pathway: Intensive PK sampling pathway aligned with Phase 1 oncology requirements **Education Protocol (HCP):** Protocol-specific training for complex fixed-sequence dosing and PK washout periods **Education Protocol (Patient):** Informed consent outlining rigid onsite requirements and strict prohibited medication rules (e.g., anti-acids, CNS depressants) **Quality Audit Mechanism:** Real-time PK sample integrity tracking and dosing adherence audits

14. Observation & Follow-Up Schedule

Total Duration: Approximately 6 weeks (from Screening through Follow-up) **Onsite Visit Cadence:** Visit 1 (Screening), Visit 2 (Period 1), Visit 3 (Period 2), Visit 4 (Period 3), and Visit 5 (Follow-up) **Remote Monitoring Frequency:** N/A (Intensive Phase 1 continuous safety oversight) **Checklist Review Interval:** Prior to each fixed-sequence dosing period (post ≥ 48 hr washout)

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				